

Recover the central spectral copy

Background

For a signal band-limited to $|w| \leq W$ and sampling angular frequency $ws > 2W$, an ideal reconstruction filter has gain $T=2\pi/ws$ in its passband. Its cutoff must satisfy $W < wc < ws-W$. Set $H(w)=T$ for $|w| < wc$ and zero elsewhere, including the cutoff points.

Your task

Determine whether a proposed cutoff is valid. Construct H and return the recovered spectrum $X_r=X_p H$. Use the supplied sampled triangular spectrum to compare two valid cutoffs and one invalid cutoff. Reject an invalid choice by returning `None`.

Function contract and assumptions

`valid_cutoff(W, ws, wc)` returns a Boolean. `recover_spectrum(w, xp, W, ws, wc)` returns (H, x_r) if valid, otherwise `None`. w and x_p are matching one-dimensional arrays; W and ws are positive. The supplied helper already creates X_p .

Example / correctness check

For $W=10$ and $ws=30$, $wc=12$ and $wc=15$ are valid; $wc=10$ and $wc=20$ are invalid. With a valid cutoff, $\max |X_r-X|$ should be below $1e-10$ on the supplied grid.

Exam instructions

Complete only the TODO functions in `template.py` and preserve their signatures. The supplied driver handles printing and plotting. Your functions must work for all valid inputs described above, not just the example. No external files or packages beyond NumPy and Matplotlib are needed. Small floating-point differences are acceptable.

Topic reference: PDF pages 24-26.